

Dhiraj Rishi Atmakuri

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Computer Engineer (May 2026) spanning embedded systems and full-stack — from BLE firmware and circuit bring-up to React Native apps and cloud backends.

EDUCATION

Texas A&M University — College Station, TX
Bachelor of Science in Computer Engineering

Aug 2022 – May 2026

PROJECTS

Water Reflection Renderer Computational Photography (CSCE 448)

Spring 2026

Python · OpenCV · NumPy

- Built an interactive Python/OpenCV pipeline generating realistic water reflections with seven real-time parameters and a live preview.
- Implemented perspective warp, sine-ripple displacement, Gaussian blur, fade/darkening, and seam-blended compositing with dirty-flag caching.

BluStick (MACTrack 5) — Real-Time BLE Tracking System Senior Capstone (CSCE 483)

Aug – Dec 2025

ESP32 · React Native (Expo) · BLE (GATT) · Firebase Auth · Cloud SQL (PostgreSQL)

[Portfolio ↗](#) · [Code ↗](#)

- Built the full Android app (React Native/Expo, TypeScript) — Firebase Auth, a live map, and logging screens — backed by a Cloud SQL PostgreSQL database on Google Cloud.
- Wrote ESP32 NimBLE firmware (C): GAP scanning into a per-MAC hashmap, Kalman-filtered RSSI-to-distance, and a GATT notify/whitelist server.
- Shipped a \$69.57 belt-worn device meeting all 10 specs: 24-hour battery, 15 m range, ~10% distance error.

Human Radar — Autonomous Elder-Care Robot Embedded Systems (CSCE 462)

Aug – Dec 2024

Raspberry Pi 5 · MediaPipe · OpenCV · Python · LiDAR · PID · GPIO [Code ↗](#)

- Designed a MediaPipe fall-detection algorithm flagging falls when the head holds below a ground threshold for 5+ seconds.
- Built autonomous person-following with a PID pan/tilt camera and LiDAR; added CLAHE/gamma correction and NoIR night vision for low light.
- Wired and soldered the Raspberry Pi 5, L298N driver, and dual power supplies; debugged a faulty driver and a USB-C power fault.

Panda Express POS — Full-Stack Web Application Software Engineering (CSCE 331)

Fall 2024

Next.js · React · Node.js · PostgreSQL · REST APIs · Agile

- Built a full-stack POS platform (Next.js/React, Node.js, PostgreSQL) with four role-based interfaces and real-time inventory tracking.
- Integrated auth, translation, and weather APIs; led as Scrum Master across three Agile sprints.

EXPERIENCE

Undergraduate Research Assistant — Texas A&M University

May – Aug 2025

- Trained deep-learning models to classify agricultural seed images; improved accuracy by tuning architecture and evaluating metrics.
- Built a UI for the AI seed-classification research pipeline.

TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript, JavaScript, SQL, Verilog

Embedded & Hardware: ESP32, Raspberry Pi, BLE (GAP/GATT), GPIO, PWM, LiDAR, sensor integration, power regulation, soldering

Web & Mobile: React, React Native (Expo), Next.js, Node.js, REST API design

Tools & ML: Git, PostgreSQL, Firebase, Google Cloud, Linux, Jira, Agile/Scrum, OpenCV, MediaPipe

ACTIVITIES

Involvement: Aggie Coding Club · Tennis Summer Coaching